Koustav Chandra

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Research Interests

Gravitational-Wave Data Analysis — Development of modelled and machine-learning searches for binary black hole mergers, parameterised tests of General Relativity, black hole spectroscopy, likelihood acceleration and inference methods, and forecasting studies for next-generation detectors (Cosmic Explorer, Einstein Telescope).

Education

Indian Institute of Technology, Bombay — Mumbai, IN
Ph.D. in Physics (Supervisor: Prof Archana Pai) — 2018–2023

- Thesis: "Hearing gravitational waves from intermediate-mass black hole binaries: Detection and Characterisation."

National Institute of Technology, Rourkela — Rourkela, IN
Integrated M.Sc. in Physics and Astronomy — 2013–2018

- Thesis: "An Algebraic Study of SO(10) Grand Unified Theory"

Experience

Max-Planck-Institut für Gravitationsphysik, Potsdam — Potsdam, DE
Postdoctoral Scholar — 2025–Current

Institute for Gravitation & the Cosmos, The Pennsylvania State University — State College, US
Postdoctoral Scholar — 2023–2025

Mentoring

- Mr Leo Ng (Penn State University) — 2026 –Current
  PhD candidate. Co-supervising a project on a modelled binary black hole search.
- Mr Giovanni Benetti (Università di Padova) — 2025–2026
  Master's student. Supervised a project on the impact of systematics on golden dark siren cosmology. Preprint: arXiv: 2602.14898
- Ms Akshita Mittal (Gran Sasso Science Institute, Italy) — 2024–Current
  PhD student (initially mentored as an undergraduate). Supervised a project assessing the impact of eccentricity on full and ringdown-only analysis. Resulting preprint: arXiv: 2606.15150
- Ms Shiksha Pandey (Penn State University) — 2024
  Undergraduate research assistant. Co-supervised project assessing the role of LIGO-India in multi-messenger astronomy for next-generation detectors. Resulted in publication: Astrophys. J. Lett. 985 (2025) L17.
- Mr Sayantan Ghosh (IIT Bombay) — 2023–2024
  PhD student. Developed coherence tests to distinguish noise transients from intermediate-mass black hole binary signals by assessing posterior consistency across detectors. Resulted in publication: Phys. Rev. D 109, 064015 (2024)
- Ms Kritti Sharma (IIT Bombay) — 2022–2023
  Undergraduate student. Developed a transfer learning framework to detect mass-asymmetric black hole binaries in Advanced LIGO data. Resulted in publication: Phys. Rev. D 108 124061 (2023)
- Mr Sagar Gupta (IIT Bombay) — 2022–2023
  Undergraduate student. Implemented a chirp-mass–based classifier for glitch identification in gravitational-wave data, improving transient search data-quality vetoes.

Invited Talks, Lectures & Seminars

♮ **Selected topics from the Compact Binary Gravitational Wave landscape** — April 2026
IMPRS Lecture Week "Gravity at the Extreme: from Theory to Observation", Wandlitz, Germany

♠ **The coming flood of Gravitational Waves in the Mid 2030s: Myths, Forecasts and Surprises** — March 2026
Instituto Galego de Física de Altas Enerxías, Universidade de Santiago de Compostela, Spain

- ♠ **GW231123: The Shape-Shifting Gollum** August 2025
  Presidency University, Kolkata & Jawaharlal Nehru University, Delhi, India
- ♠ **Too Many, Too Long, Too Loud: CBC Data Analysis Challenges in the XG Era** May 2025
  PAX-X and CE Symposium, University of Illinois Urbana-Champaign, USA
- • **Exploring the XG Era: Foreground Noise Impact & LIGO India's Role** November 2024
  Detection and Analysis of Gravitational Waves in the Era of Multi-Messenger Astronomy, Banff, Canada
- ♠ **Preparing for a Gravitationally-Bright Tomorrow** August 2024
  Department of Physics, Indian Institute of Technology Bombay, Mumbai, India
- • **Intermediate-Mass Binary Black Holes and How to Find Them** February 2023
  Inference-Data-Experiments-Analysis (IDEA) Talks, TIFR, Mumbai, India
- • **Prospects of Observing Mass-Asymmetric IMBH Binaries with A+ Sensitivity** May 2023
  Science with LIGO-India Meeting, IUCAA, Pune, India
- • **Hunting for IMBHs with the International Gravitational-Wave Observatory Network** February 2022
  Chennai Symposium on Gravitation and Cosmology, Chennai, India

♮ Invited Lecture ♠ Invited Seminar • Invited Talk

Academic Awards & Fellowships

- • **Naik and Rastogi Excellence in Ph. D. Thesis Award**, Indian Institute of Technology, Bombay 2024
- • **Department Prize for Outstanding Student Performance**, National Institute of Technology, Rourkela 2016

Selected Publications

For full list see: inspirehep.net/authors/1817603. ⊛ := papers co-authored with PhD supervisor

Short-Author Publications

[1] J. Calderón Bustillo, S. H. W. Leong, and **K. Chandra**, "A complete measurement of a black-hole recoil through higher-order gravitational-wave modes", *Nature Astron.*, vol. 9, no. 10, pp. 1530–1540, 2025.

[2] **K. Chandra**, "gwforge: a user-friendly package to generate gravitational-wave mock data", *Class. Quant. Grav.*, vol. 42, no. 2, p. 025 003, 2025.

[3] **K. Chandra**, R. Gamba, and D. Chiaramello, "From Source Properties to Strong-Field Tests: a multipronged analysis of GW250114 with an effective one-body model for generic orbits", Dec. 2025. arXiv: 2512.04593 [gr-qc].

[4] S. Pandey, I. Gupta, **K. Chandra**, and B. S. Sathyaprakash, "The Critical Role of LIGO-India in the Era of Next-generation Observatories", *Astrophys. J. Lett.*, vol. 985, no. 1, p. L17, 2025.

[5] A. Rashti, R. Gamba, **K. Chandra**, D. Radice, B. Daszuta, W. Cook, and S. Bernuzzi, "Binary black hole waveforms from high-resolution gr-athena++ simulations", *Phys. Rev. D*, vol. 111, no. 10, p. 104 078, 2025. arXiv: 2411.11989 [gr-qc].

[6] **K. Chandra**, I. Gupta, R. Gamba, R. Kashyap, D. Chattopadhyay, A. Gonzalez, S. Bernuzzi, and B. S. Sathyaprakash, "On the Origins, Remnant, and Multimessenger Prospects of the Compact Binary Merger GW230529", *Astrophys. J.*, vol. 977, no. 2, p. 167, 2024. arXiv: 2405.03841 [astro-ph.HE].

[7] **K. Chandra**, J. Calderón Bustillo, A. Pai, and I. W. Harry, "First gravitational-wave search for intermediate-mass black hole mergers with higher-order harmonics", *Phys. Rev. D*, vol. 106, no. 12, p. 123 003, 2022, ⊛.

[8] **K. Chandra**, V. Villa-Ortega, T. Dent, C. McIsaac, A. Pai, I. W. Harry, G. S. C. Davies, and K. Soni, "An optimized PyCBC search for gravitational waves from intermediate-mass black hole mergers", *Phys. Rev. D*, vol. 104, p. 042 004, 2021, ⊛.

LVK Collaboration Publications with Significant Contributions

[1] A. G. Abac *et al.*, "GW231123: A Binary Black Hole Merger with Total Mass 190–265 $M_\odot$", *Astrophys. J. Lett.*, vol. 993, no. 1, p. L25, 2025. eprint: 2507.08219.

[2] "GW230814: investigation of a loud gravitational-wave signal observed with a single detector", Sep. 2025. arXiv: 2509.07348 [gr-qc].

[3] R. Abbott *et al.*, "Search for intermediate-mass black hole binaries in the third observing run of Advanced LIGO and Advanced Virgo", *Astron. Astrophys.*, vol. 659, A84, 2022. eprint: 2105.15120.

[4] R. Abbott *et al.*, "GW190521: A Binary Black Hole Merger with a Total Mass of $150M_{\odot}$", *Phys. Rev. Lett.*, vol. 125, no. 10, p. 101 102, 2020. arXiv: 2009.01075 [gr-qc].